

● MEDIUM

● ADVANCED MATH

Equivalent expressions

10 questions · ~15 min

01

Q1

ADVANCED MATH

Which of the following is equivalent to $\sqrt[4]{x^2+8x+16}$, where $x > 0$?

A. $(x+4)^4$

B. $(x+4)^2$

C. $(x+4)$

D. $(x+4)^{\frac{1}{2}}$

Q2

GRID-IN

ADVANCED MATH

$$2x^2 + 5x - 12$$

If the given expression is rewritten in the form $(2x - 3)(x + k)$, where k is a constant, what is the value of k ?

STUDENT-PRODUCED RESPONSE

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.	/	.	/	.	.
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Which expression is equivalent to $\frac{h^{15}q^7}{h^5q^{21}}$, where $h > 0$ and $q > 0$?

A. $\frac{h^{10}}{q^{14}}$

B. $\frac{h^3}{q^3}$

C. $h^{10}q^{14}$

D. h^3q^3

Q4

GRID-IN

ADVANCED MATH

$$(5x^3 - 3) - (-4x^3 + 8)$$

The given expression is equivalent to $bx^3 - 11$, where b is a constant. What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

ADVANCED MATH

If $a = 4k + 5r$ and $b = 7k - 12r + 3$, which expression is equivalent to $a - b$?

- A. $-3k + 17r + 3$
- B. $-3k + 17r - 3$
- C. $-3k - 7r - 3$
- D. $-3k - 7r + 3$

Q6

ADVANCED MATH

Which expression is equivalent to $8x^3 + 8 - x^3 - 2$?

- A. $8x^3 + 6$
- B. $7x^3 + 10$
- C. $8x^3 + 10$
- D. $7x^3 + 6$

Q7

ADVANCED MATH

$$f(x) = x^2 + bx$$

$$g(x) = 9x^2 - 27x$$

Functions f and g are given, and in function f , b is a constant. If

$f(x) \cdot g(x) = 9x^4 - 26x^3 - 3x^2$, what is the value of b ?

A. -26

B. $-\frac{26}{9}$

C. $\frac{1}{9}$

D. 9

Q8

ADVANCED MATH

Which expression is equivalent to $6x^8y^2 + 12x^2y^2$?

A. $6x^2y^22x^6$

B. $6x^2y^2x^4$

C. $6x^2y^2x^6 + 2$

D. $6x^2y^2x^4 + 2$

Q9

ADVANCED MATH

Which of the following expressions is equivalent to $8x^{10} - 8x^9 + 88x$?

A. $x7x^{10} - 7x^9 + 87x$

B. $x8^{10} - 8^9 + 88$

C. $8xx^{10} - x^9 + 11x$

D. $8xx^9 - x^8 + 11$

Q10

ADVANCED MATH

Which of the following is equivalent to $(1 - p)(1 + p + p^2 + p^3 + p^4 + p^5 + p^6)$?

A. $1 - p^8$

B. $1 - p^7$

C. $1 - p^6$

D. $1 - p^5$